

Generative AI (Gen AI) – One Page Syllabus

Module 1: Foundations of AI & ML

AI vs ML vs Deep Learning, supervised/unsupervised learning, neural networks basics.

Module 2: Large Language Models (LLMs)

Transformers, attention mechanism, tokenization, embeddings, GPT, Llama, Gemini concepts.

Module 3: Prompt Engineering

Zero-shot, one-shot, few-shot prompting, chain-of-thought, structured prompts, evaluation.

Module 4: Retrieval-Augmented Generation (RAG)

Embeddings, vector databases (ChromaDB, Pinecone), similarity search, retrieval pipelines.

Module 5: Gen AI Application Development

LangChain, agents, tools, memory, chatbots, document Q&A; systems.

Module 6: Fine-Tuning & Model Optimization

Fine-tuning concepts, LoRA, PEFT, quantization, inference optimization.

Module 7: Deployment & MLOps

API integration, Streamlit, FastAPI, Docker, cloud deployment, monitoring.

Module 8: Responsible AI

Bias, fairness, privacy, security, hallucinations, ethics and governance.

Hands-On Projects

AI Summarizer, RAG Chatbot, Placement Preparation Assistant, Exam Generator, AI Agents.